

```
import cv2
from google.colab.patches import cv2_imshow

# Read image in grayscale
img = cv2.imread("lab_image.jpg", 0)

# Check if image is loaded
if img is None:
    print("Image not found!")
else:
    # Sobel along X-axis
    sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)

    # Sobel along Y-axis
    sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

    # Combine X and Y edges
    edges = cv2.addWeighted(sobelx, 0.5, sobely, 0.5, 0)

    # Convert to unsigned 8-bit image
    edges = cv2.convertScaleAbs(edges)

    # Save output image
    cv2.imwrite("Edge_detection.jpg", edges)

    # Display images
    print("Original Image")
    cv2_imshow(img)

    print("Sobel XY Edge Detection")
    cv2_imshow(edges)
```

Original Image



Sobel XY Edge Detection



